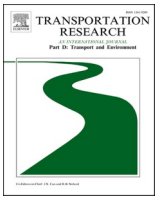
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Study of channel upgrades and ship choices of river-shipping of port access-transportation

Yu Sun, Zhouting Lu, Feng Lian, Zhongzhen Yang *

Donghai Academy, Ningbo University, Ningbo, China

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ABSTRACT

This study builds a bi-level programming model to optimize river channel upgrades and liner operation, while minimizing the total cost of container transportation to ports. The upper model determines the optimal channel upgrades, choice of liner size, and liner operating frequency. The sum of the trucking freight, the river liner shipping cost, the CO_2 tax payment, and the depreciation of the investments in channel upgrades and ship purchases is calculated based on official projections of freight demand over 30 years. The lower model determines the optimal container routes. A case study of Zhejiang Province shows that Qiantang River and Hangzhou-Ningbo Canal should be upgraded from Level IV to Level III waterways, while the Meihu and Zhajiasu Lines should be upgraded from Level V to Level IV waterways. Due to these improvements, the mode share of river shipping is projected to increase to 30%, while CO_2 emissions from container port access transportation are estimated to decrease by 23.5 million tons over 30 years.

1. Introduction

Due to its status as a major manufacturer, the value of China's total import and export volume reached US$6.3 trillion in 2022. The continuous growth of international trade with China over the prior 30 years has produced a correspondingly significant growth in freight demand at Chinese ports. As a result, Chinese port container throughput has reached new year-on-year highs since 2000. Fig. 1 below depicts the volume of port container throughput from 2016 to 2020.

The vast majority of these containers are transported in or out of ports by trucks, which consume large quantities of fuel and emit a high volume of CO_2. In order to control CO_2 emissions, China has recently redoubled its efforts to promote modal shifts in container port access and egress transportation, primarily via economic and administrative measures aimed at increasing the mode share of railway transportation. Several such measures include the normalization of container liner trains at ports (namely, regularly scheduled access and egress container trains), increases in the number and coverage of extant liner trains, the construction of on-dock or near-dock railway terminals and inland dry ports, reductions in freight rates for liner trains, and the subsidization of sea-rail intermodal pilot projects. In addition to encouraging the modal shift from highways to railways, China has also vigorously developed its river shipping for container port access and egress transportation such that several river ports along the golden waterway of the Yangtze River have received continual upgrades to their service capacity. Simultaneously, river liner capacity between the river ports and Shanghai has been expanded with the aim of increasing the utility of the freight chain composed of short-distance trucking to river ports, medium-distance Yangtze River liner shipping to gateway ports, and finally, ocean liner shipping to overseas destinations. These

* Corresponding author.
E-mail address: yangzhongzhen@nbu.edu.cn (Z. Yang).

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upgrades have also tended to increase river shipping's modal share of container port access and egress transportation in the Yangtze River Economic Belt.

Although these incentives to change the mode shares of container port access transportation were implemented with the goal of reaching carbon peaking and neutrality targets, the shares of railway and river shipping in port access transportation have remained stagnant at 2% and 10%, respectively (Tao and Wu, 2021). Even in regions that enjoy good river channels, river shipping as a mode of port access transportation continues to grow only slowly. The main reasons for this are, first, that in regions rich in river channels, export containers frequently originate close to gateway seaports (usually within 100–500 km), allowing the timeliness of trucking to fully compensate for its economic disadvantages. Second, with the exception of the Yangtze River, the capacity of most other inland river channels remains insufficient, and problems related to poor locks cause bottlenecks in river shipping. Third, river shipping routes lack systematic design, and the ships, channel construction, and placement of key choke nodes resist integration into a coordinated network. Finally, a carbon tax has not yet been imposed within China. Without considering this proposed carbon emission cost, short-range river shipping has no advantages over trucking.

Given this background, for the sake of our study, it is necessary to select a sample province that is rich in river channels but not located along the Yangtze River in order to clarify the flow direction and volume of export containers, as well as the shippers' mode choice preferences. It is also necessary to analyze the bottlenecks so prevalent along inland waterways, and to optimize the river shipping network in order to avoid this problem. Thus, this study proposes a bi-level programing model that aims to minimize the sum of the carbon tax payment, the investment in river channel upgrades and ship purchases, and the costs in both time and money required to transport containers from their origins to the destined gateway seaport. The upper model determines the optimal upgrades to river channels, choice of river liners, and their operating frequency based on the projected freight demand over the next 30 years. The lower model calculates the probabilities of shippers choosing a given route, and identifies the likeliest container travel paths. Based on the outputs from both the upper and lower models, we then calculate the total cost (including trucking freight payment, river shipping cost, CO_2 tax payment, and the depreciation value of the investments in channel upgrades and ship purchases). For the case study, this paper designs a model solution algorithm. Based on the numerical results from the case study, we recommend an optimized scheme of river channel upgrades, new ships, liner operating frequency, and a method of eliminating several bottlenecks. These results lay the theoretical basis for future increases in the mode share of river shipping in container port access transportation, both in the sample province and in similar regions.

The remainder of this paper is structured as follows: Section 2 conducts the literature review, and Section 3 presents the problem description. Section 4 develops the model's construction, Section 5 conducts a case analysis, and Section 6 summarizes the study and its significant implications.

2. Literature review

In the realm of container transportation, studies on energy saving and emission reduction focus primarily on four aspects, namely, sailing speed optimization and alternative fuels, modal shifts toward inland waterway transport, assessments of environmental costs, and policy suggestions.

The carbon emissions from liner shipping depend not only on a ship's sailing speed, but also on the marine fuel used. Wang and Meng (2012) calibrated the curve of fuel consumption and sailing speeds for container ships by using historical liner operation data. They then optimized the sailing speed along each route of a liner shipping network with the aim of minimizing bunker consumption

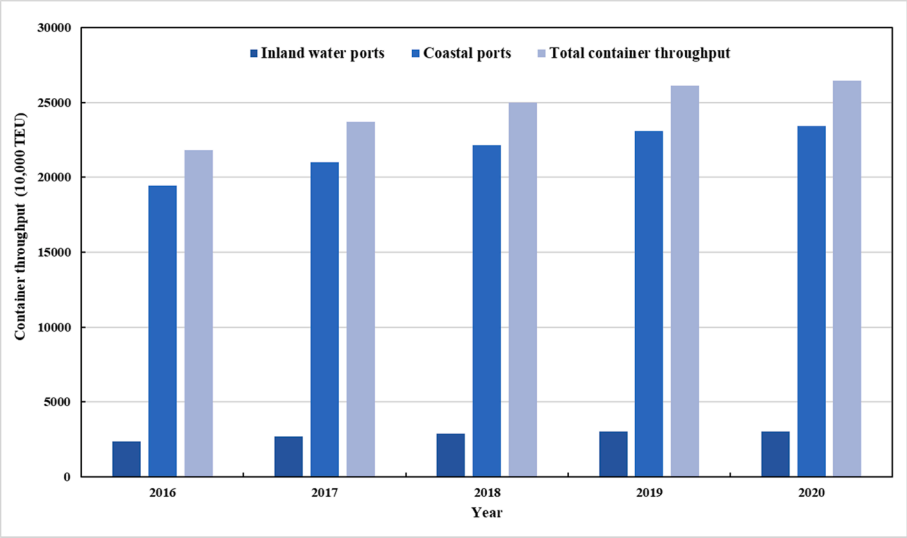


Fig. 1. Port container throughput in China, 2016–2020 (Source: China Port Statistical Yearbooks).

and providing a practical approach to the slow steaming strategy frequently used to reduce ships' operating costs and CO_2 emissions. Chang and Wang (2014) established a model of slow steaming and emission reduction so as to maximize the cost-benefit tradeoffs from slow steaming. Their analysis demonstrated that when the charter rate of a ship is low and the bunker price is high, the ship's operating cost and CO_2 emissions can be effectively reduced via slow steaming. Psaraftis and Kontovas (2014) constructed a sailing speed optimization model to minimize the total cost of bunker fuel, charter fees, and cargo inventory by considering the influence of ship load on bunker consumption. The optimized results showed that the speed that produces the lowest level of emission is not the most economical speed, and that a fuel tax would likely encourage liners to lower their sailing speeds and thus reduce both bunker consumption and CO_2 emissions. Simultaneously, they found that the operating costs of the liner companies would increase, as more ships would be needed. In order to study the impact of liner ships' sailing speeds on operators, shippers, and the environment from a supply chain perspective, Tran and Lam (2022) developed a model to simulate the transatlantic container flows of liner operators at different sailing speeds. They found that sailing speed has little impact on the total cost to the supply chain, but has a large impact on cargo turnaround times and carbon footprint. Further, they demonstrated that slow steaming reduces bunker consumption and CO_2 emissions from ship engines, but lengthens the transit time of cargo, which increases the inventory cost of a shipper's cargo. This section of the literature is focused on ocean liners and their slow steaming, with a particular focus on an existing liner fleet and a single liner route's one-time point OD freight demand. The studies conducted in this area rely on fixed values for carbon tax, liner fleet, and freight demand, which limit their ability to account for future scenarios. Our study aims to address this limitation by optimizing the operation scheme of several inland river liner routes based on the projections of multi-time point OD freight demands, while considering the fleet life cycle, the long-term systematic nature of liner shipping service, and with the objective of minimizing the total cost of multi-modal cooperative container port-access transportation.

As new energy sources come into widespread use, the literature on the replacement of traditional fuels with these clean fuels is also increasing. Tai and Lin (2016) simulated the redeployment of ocean trunk routes globally after the expansion of the Panama Canal, given a variety of approaches to shipping operations, and proposed a clean-line strategy to replace fuel oil with liquefied natural gas. They then tested the effects of this strategy on energy savings and emission reductions for liner ships, and demonstrated that the adoption of these new measures along the redeployed eastbound trunk route between the Far East and Europe could significantly reduce the carbon emissions from liner shipping. In order to reduce the carbon footprint and life cycle costs of inland river shipping, Perčić et al. (2021) proposed a number of emission reduction plans to replace traditional bunker fuels with electricity, methanol, liquefied natural gas, hydrogen, ammonia, and biodiesel, after which they established a carbon footprint model for inland river vessels. Furthermore, they evaluated these alternative bunker fuel options by applying the life cycle cost assessment method to ships operated in Croatia. They found that ship electrification offered the most energy savings and reductions in emissions, and that electrification reduced the carbon footprint of ships by 51%. Ye et al. (2022) designed an electrical power system for water taxis and container ships powered by hydrogen polymer electrolyte membrane fuel cell (PEMFC) systems. Through a comparison with traditional fuel engines, they found that gaseous hydrogen PEMFC, liquid hydrogen PEMFC, and liquid ammonia PEMFC power systems reduced carbon emissions by 91%, 91% and 88%, respectively. In the absence of a carbon tax, ship operating costs rose by 108% (gaseous hydrogen), 112% (liquid hydrogen), and 116% (liquid ammonia) over a 25-year life cycle; meanwhile, the cost of water taxis rose by 43% (gaseous hydrogen) and 105% (liquid ammonia). When the carbon tax was set at 75–191 GBP per ton of CO_2, the new system demonstrated a cost advantage. Although many studies have found that optimizing vessel speeds and using alternative fuels can reduce ships' carbon footprints, they have also demonstrated that such strategies tend to increase the total operating costs to shipping companies. This section of the literature analyzes the impact of bunker fuel replacement on emission reduction throughout a ship's life cycle. However, the calculations were performed solely for a single ship under the assumption that the onboard cargo remains constant. The study did not consider market equilibrium factors, such as competition from truck transportation, which could affect the efficacy of switching bunker fuel for a ship.

Due to the economic and environmental advantages of inland waterway transport (Tournaye et al., 2010), many countries are actively developing this mode of transportation. In the past 20 years, the European Union has implemented a number of development and funding programs targeted at inland waterway transportation, including Marco Polo, Motorways of the Sea, the Trans-European Transport Network (TEN-T), the Navigation and Inland Waterway Action and Development in Europe (NAIADES) program, and the EU Interregional Cross-Border Cooperation (INTERREG) (Baird, 2007; Rogerson et al., 2020; Solomon et al., 2021; Hartz and Caesar, 2022). Kotowska et al. (2018) analyzed and categorized the measures implemented by the ports of Antwerp, Rotterdam, and Hamburg to promote inland waterway transportation in their environs. They found that these actions were quite similar in terms of their scope. The most common external actions included initiatives that improved the quality of inland waterways and the quality of shipping connections with the hinterland. Internal initiatives largely focused on improving the port's infrastructure for barges, and on implementing an appropriate information policy. Rogerson et al. (2020) drew on two cases in which Swedish entrepreneurs initiated inland waterway transport development projects to study how barriers to a modal shift toward river shipping can be managed where markets are limited. They found that such a modal shift involves many stakeholders, and outlined the stakeholders necessary for successfully overcoming barriers to development, stressing the particular roles of different actors. In particular, they suggested that policy makers aiming to promote such a modal shift should clearly identify regulatory barriers and their own role in facilitating the desired modal shift through the reduction of such barriers. In the context of the Indian subcontinent, Hasan et al. (2021) analyzed the intermodal freight diversion from highway to inland waterway transport in Bangladesh. Specifically, they used the inventory theory model to compare the total logistics costs of highway, railway, and inland waterway transportation along the Dhaka-Chittagong corridor, and found that river shipping was cheaper than highway transportation, although it resulted in longer transit times. Trivedi et al. (2021) employed Decision-Making Trial and Evaluation Laboratory (DEMATEL) and Interpretative Structural Modeling (ISM) methods to analyze the effects of regulatory barriers on the implementation of inland water transport. They found that governance

issues, policy bias, high cost requirements, and a lack of river-interlinking are the most critical inhibitors of inland water transportation development in India. This section of the literature examines policy initiatives aimed at promoting modal shift, including improving inland waterways, providing subsidies, and eliminating policy bottlenecks. However, in this paper, we not only cover these elements but also provide additional feasible policy measures to promote modal shift. We conduct a scenario analysis for freight demands at various time points and different carbon tax prices, while considering market equilibrium, to offer a detailed basis for stakeholders to put forth policies and measures aimed at reducing carbon emissions from container port-access transportation. Such policies and measures may include specific inland waterway upgrade schemes, determining the appropriate ships for each inland waterway route, and identifying the carbon tax price required to achieve different carbon reduction goals.

Following the experience of inland waterway transportation development in Europe, the Asian Development Bank compiled their 2016 report, *Promoting Inland Waterway Transport in the People's Republic of China.* This report put forward suggestions for improving port infrastructure construction, boosting the service quality of inland waterway transport, strengthening the relationship between the government and state-owned enterprises, and enhancing the participation of stakeholders in inland waterway transport. Wu and Zhang (2022) further analyzed the freight network of China's Yangtze River Economic Belt and found that measures including a carbon tax, an intermodal transport subsidy, and other capacity expansion policies would likely help to promote a modal shift from highway to railway and inland water transportation. The studies and reports reviewed here have analyzed the obstacles to a modal shift in favor of inland waterway transportation at a macro level and suggested methods of overcoming major regulatory hurdles. So far missing from the discussion, however, is any proposed optimization plan for increasing river shipping's mode share of container port access and egress transportation (thereby reducing carbon emissions around ports) by improving the infrastructure and transport ships necessary to river transportation.

In the context of policy objectives related to carbon neutrality and curbing emissions, the potential cost savings from emission controls have become a driving force for enterprises seeking to reduce CO_2 emissions. By taking carbon costs into account, Chang et al. (2014) established a bi-objective inland dry port location model that sought to minimize the total generalized cost. They found that the carbon emitted by inland transportation demonstrated an inverse relationship to the number of inland dry ports and the total cost of inland transportation. Lam and Gu (2016) proposed a bi-objective multimodal transportation network optimization model with carbon footprint as the constraint, and the minimum total operating cost and transportation time as the two objectives. They then used the model to quantify the impacts of changes in the limit of a unit container's carbon emission on the multimodal transportation network. Based on multi-commodity pickup and delivery rules, Cariou et al. (2018) proposed a mixed-integer linear programming model with which to optimize ports of call, the order of the calls, container flows between ports, and ship allocation by considering the costs of carbon and sulfur emissions. Yu et al. (2021) studied the problem of profit maximization from maritime transportation, and proposed a flexible route optimization model based on the scheduled sailing period that would identify a route network capable of improving shipping margins and reducing carbon costs. Their calculations showed that the optimal routes increased the operating income of liner companies by 3.7% while reducing carbon emission costs by 33.47%. Wang et al. (2022a) established a mixed-integer nonlinear programming model to minimize the total costs of CO_2 emissions, fuel consumption, port charges, operation costs, and shippers' inventory (subject to the load capacity of ships) in order to determine the optimal voyage frequency and vessel size for liner shipping. Their results demonstrated that, as the size of the liner ship increased, the total cost first decreased and then increased, while the carbon emission cost as a proportion of the total costs gradually increased. These studies all take environmental factors as constraints or targets in the optimization of a network, but most leave aside any examination of the effects of carbon costs on impedance between transportation modes and resistance to shifts in mode shares. Furthermore, the focus of these studies is on ocean liners, where the composition of shipping routes and their competing modes are relatively straightforward. Therefore, it was unnecessary for them to consider the trip-end transportation of multimodal transportation.

In order to control the greenhouse effect, governments around the world have successively introduced measures to reduce carbon emissions. Researchers have devoted considerable attention to the impact of these measures on carbon emissions in container transportation. Bickford et al. (2014) used the Wisconsin Inventory of Freight Emissions and the Community Multiscale Air Quality Model to quantify the impact of the modal shift from trucking to rail transportation on emissions. Their investigation found that such a modal shift in the midwestern United States could reduce CO_2 emissions by 23 million tons per year, which is equivalent to the reduction of emissions from 4 million passenger cars. Wang and Xu (2015) established a sailing speed decision model in order to analyze the impact of different carbon tax approaches (namely, no tax, a tax on carbon emissions over a specified threshold, and a tax on all emissions of CO_2) on the operating profits of carriers and on ships' carbon emissions. The model took the bunker cost, carbon tax cost, and demurrage charges into account and found that any tax on CO_2 emissions reduced the profits of carriers and ship emissions simultaneously, although the tax on emissions beyond a certain limit produced better results in terms of emissions reduction and revenue effects. Xing et al. (2019) established a mixed-integer nonlinear programming model in order to optimize ship sailing speeds and fleet deployment under two different policies, namely, a cap on carbon emissions or a carbon tax. Lan et al. (2020) established a liner shipping optimization model under four different carbon emission policies (no emission limit, an emission tax, an emission cap, and an emission trading exchange) so as to study the impact of different control methods on the operating profits of carriers and volume of emissions. They found that mandatory carbon taxes were effective, while a carbon cap-and-trade system had only a weak effect on profits and emissions. Ding et al. (2020) examined the impact of two types of carbon taxes (that is, a fixed charge and a progressive tax) on the economic efficiency of Arctic shipping. Their study demonstrated that, while each form of taxation increased ship operating costs and reduced shipping companies' profitability, the additional costs of a progressive carbon tax were less than those imposed by a fixed carbon tax, making a progressive carbon tax more acceptable. Li and Zhang (2020) proposed a profit maximization model of carbon taxation that considered internal measures (such as dynamic pricing and flexible scheduling) and external policy subsidies aimed at promoting a shift from trucking to rail freight. Their results showed that dynamic carbon tax pricing reduced the

volume of CO_2 emitted from the transportation network by 3,261 tons. Simultaneously, a subsidy of 0.1–0.6 CNY per TEU per km for rail freight achieved the purpose of transferring 23.78–25.38% of cargo from roads to railways, reducing CO_2 emissions by 24.08–26.12%. Zhang et al. (2021) used a transportation dataset with high spatial resolution to simulate the traffic flow changes caused by modal shifts under three strategies (namely, a combined port-railway strategy, a combined port-railway-waterway strategy, and a policy-enhanced strategy). They then adopted the emission factor model to establish an emission inventory, which they then used as an input for the air quality model in order to forecast changes in the concentration of major air pollutants and evaluate the environmental impact of Shenzhen's multimodal transport plans. Lagouvardou et al. (2022) studied the effect of fuel taxes on ship emissions reduction in the short term. Based on data related to ship sailing speeds, bunker prices, and the spot exchange rate, they analyzed the relationship between bunker prices, freight rates, and ship sailing speeds from 2010 to 2018 and established a tanker speed optimization model for both full and ballast states. Although studies such as those surveyed here have deeply analyzed the effects of carbon taxes on emissions, the precise impact of carbon tax prices on the operating costs to shipping companies and on shippers' preference of freight mode demands further clarification. Additionally, some of their quantitative studies assumed that ship assets and freight demand were deterministic, and then analyzed the impact of carbon tax on the operating costs of ships and carbon emissions from cargo transportation. In contrast, in this paper we set up several dynamic scenarios, including multiple scenarios of freight demand and market competition scenarios between trucks and river liners, to provide a more comprehensive analysis.

Whereas the four branches of literature examined above have studied emission reductions in container transportation at multiple levels, few of them have explored the technical aspects of realizing modal shifts to low-carbon transportation from the perspective of construction costs amortized over the life span of improved infrastructure, the monetary and time costs of transportation, and the carbon emission costs due to tax payment. In addition, all of these studies are based on relatively static scenarios for computational analysis and lack consideration of the long-term and systematic nature of liner shipping services. This context frames the major contributions of this study. For provinces with small developed inland waterways, we first optimize a scheme of upgrades to the river infrastructure and operations based on projected freight demand over the next 30 years, with the goal of minimizing the total cost of container port access transportation. This is done to eliminate the bottlenecks incurred during a modal shift from trucking to river shipping, thereby increasing the mode share of river shipping in port access transportation and reducing the resulting carbon emissions. We next analyze the reasons for river shipping's persistently low mode share of container port access and egress transportation, and put forward a method to improve its competitiveness by adjusting the type and frequency of inland river liner service. This lays a theoretical foundation for the provincial government to guide inland river shipping companies in providing access and egress transportation services. Third, we take the carbon emission cost into account when reconstructing the measurement method of the impedance of container port access transportation, and use this approach to analyze the impact of carbon emissions on mode choice. Finally, we analyze the sensitivity of the carbon tax price to river shipping's mode share of container port access transportation and to the carbon emissions from port access transportation more generally, providing a theoretical basis for national efforts to control carbon emissions from container port access transportation through reasonable adjustments to carbon tax prices.

3. Problem description

River shipping enjoys advantages of economies of scale and low carbon emissions. In order to achieve these economies of scale, however, river shipping companies habitually operate large ships at relatively infrequent intervals. Because of the low operating frequency of these liners, river shipping has no obvious advantage in the realm of short-distance transportation. Without the lower freight rate produced by the ships' economies of scale, it is difficult for river shipping to compete with trucking for container port access transportation. In addition, apart from the golden waterway of the Yangtze River, the grades of other Chinese river channels are generally low, and the necessary locks at different elevations are in particularly poor condition. All of this raises the barrier to river shipping companies that would otherwise select large ships that would allow them to compete with trucking. Thus, the upgrade of river channels (and, especially, locks at several chokepoints) in order to enable the operation of larger ships, and the implementation of a carbon tax so as to increase the cost of trucking are two key approaches to increasing the mode share of river shipping in port access transportation. More specifically, the decision problems include the following elements: (1) the determination of the upgrade scheme for river channels (including locks); and, (2) the determination of optimal port access liner shipping service from river ports to gateway

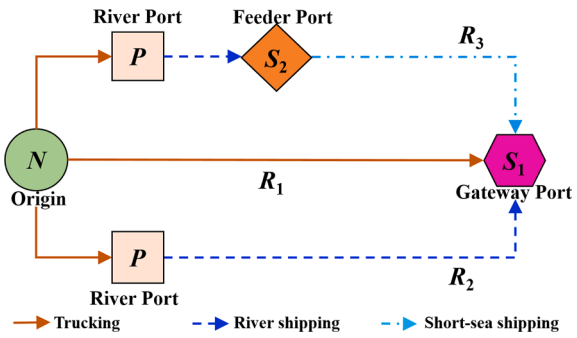


Fig. 2. Description of routes by which containers access gateway seaports.

seaports, which includes a consideration of shipping routes, the size and number of ships, and the operating frequency of river liner shipping. Since the model must also consider the competition between trucking and river shipping, and their respective payment of carbon taxes, we can reformulate the problems to include the upgrade scheme for river channels; the design of river liner shipping lines; and, the shippers' choice of route along a port access network consisting of national highways and inland waterways. The decision variables consist of the description metrics from the two decision-making problems.

Fig. 2 shows the current integrated transportation network within a Chinese province, which includes both highways and waterways. In Fig. 2, there are two types of routes possible for an export container moving from origin (N) to a gateway seaport (S_1). The highway route R_1, via direct trucking, constitutes the first path, while the two intermodal paths that feature river shipping as the main mode comprise the second type of route. Of these, R_2 is an intermodal route with both trucking and river shipping, while R_3 is a multimodal route of trucking, river shipping, and short-sea shipping. The probabilities of a shipper selecting one of these three routes determine the total cost of a given container's port access transportation, and are affected by the grades of river channels, ship size, and the operating frequency of river and short-sea shipping. In this study, we determine an upgrade plan for river channels, the optimal size of liner ships, the operating frequency of river liner shipping, and the necessary number of ships. The objective of the optimization measures proposed in this study is to minimize the sum of the costs of transporting containers to gateway seaports, the corresponding carbon tax payment, the cost of upgrading the river channels, and the cost of ship purchase and operation. For the sake of simple description, we describe the sum of these costs as the total generalized container port access cost (or total cost). In order to solve the problem, this study builds a bi-level programing model, in which the upper model provides the solutions of the decision variables and calculates all the costs, while the lower model calculates the route choice probabilities via discrete choice theory.

It is worth noting that the upper model considers the government responsible for infrastructure, while also taking into account the decisions of inland river liner companies, an analytical approach made possible by the inclusion of the cost of carbon emissions in the objective function. In the absence of a carbon tax, liner companies will not consider the cost of carbon emissions, and the liner shipping services they set may not meet socially desired targets for carbon emissions. Under the constraint of carbon emission target, however, the imposition of either carbon taxes or subsidies for low-carbon transportation renders both the government and liner companies decision makers within the upper model. The inland river liner operation plan, by considering the optimization of carbon emission target, thus provides the justification for the government to promote inland river port access and egress transportation, and to restrain truck transportation through imposing carbon taxes or providing operating subsidies.

4. Model construction

Upgrades to hydraulic infrastructure and the purchase of ships are one-time investments that provide service over many years, while the operating costs of ships and trucks can be considered a single transportation service from an origin to the destination. In order to unify the temporal scales of these different costs, we measure the costs of the investment in channel upgrades, ship purchases, ship operations, container port access transportation, and carbon tax payments over the course of one year. Given the design specifications of hydraulic structures, the life of infrastructure like river channels or locks are typically 50 years. Since these facilities may need overhaul (and thus further investment) during their life cycle, however, the industry has set the payback period for this type of infrastructure at 30 years (its economic life) (MTPRC, 2009). The financial rules of shipping companies, meanwhile, set the payback period of purchased ships at 20 years. Thus, we can calculate the average annual costs of both types of investment using the straight-line discounting method. The trucking cost is equivalent to the unit trucking price multiplied by the annual number of TEU-kilometers. Finally, river liner services usually operate on a weekly basis, such that we can use weekly container flow to estimate the weekly costs of river shipping to ports, bunker fuel, labor, and management, which we then convert into the average annual cost. Given these cost calculations, and because the upper model is designed to minimize the total generalized container port access cost over the entire economic life of river channels and locks, we can express the upper model in the bi-level programming problem as follows:

$$Min: Z = \sum_{y \in Y} C_{yB} + C_{yT} \tag{1}$$

$$C_{yB} = R_{yB} + V_{yB} + V_{yO} \tag{2}$$

$$C_{yT} = \sum_{n \in N} \sum_{r_n \in R_n} T_{y,nr_n} \left(c_{y,nr_n} + Coe_{nr_n} \times Cot \right) \tag{3}$$

In the model, Y is the payback period of river channel upgrades; C_{yB} represents the annual discounted value of the investments in river channel infrastructure improvements and ship purchases in the yth year, while C_{yT} indicates the annual container port-access costs in the yth year. In Eq. (2), R_{yB} and V_{yB} represent the respective discounted values in the yth year of the investments in river channel upgrades and ship purchases; V_{yO} is the use cost of river liner ships in the yth year. In Eq. (3), T_{y,nr_n} represents the container flow from origin n to gateway seaport through route r_n in the yth year; c_{y,nr_n} and Coe_{nr_n} give the transportation cost and the carbon emissions per container from origin n to gateway seaport via route r_n in the yth year, respectively. In addition, Cot indicates the monetary value of the carbon tax; N represents the set of origins, and $n \in N$; simultaneously, R_n is the set of feasible routes from origin n to seaport gateway, and $r_n \in R_n$.

The cost of river channel upgrades includes the investment discounting cost and the investment opportunity cost, which can be partitioned year by year over the entire period of evaluation, as shown below:

$$R_{yB} = \theta_1 T_B + T_B(1+\alpha_1)^{y-1}\alpha_1 \tag{4}$$

$$T_B = I_{channel} \times R_{channel} \times L_{channel} + I_{lock} \times R_{lock} \times N_{lock} \tag{5}$$

Here, θ_1 and α_1 indicate the annual discount rate of channel infrastructure assets and the bank's annual interest rate, respectively; T_B is the investment amount required for the upgrades. In Eq. (5), $L_{channel}$ and $I_{channel}$ give the length of the upgraded channels and the investment amount per unit of length; N_{lock} and I_{lock} are the number of upgraded locks and the investment in a single lock, while $R_{channel}$ and R_{lock} represent the optimal grades of both river channels and locks.

The cost of a ship purchase is the product of the ship's unit price and the number of purchased ships, which can also be allocated annually, as shown below:

$$V_{yB} = \theta_2\left(\sum_{i \in I_{type}} I_i \times N_{yi}\right) \tag{6}$$

Here, θ_2 is the annual discount rate of ship assets; I_{type} is the set of ship sizes, and $I_i \in I_{type}$; N_{yi} is the number of ships of size i purchased in the yth year. The use cost of river liners includes labor, bunker, and management costs, which can be expressed as follows:

$$V_{yO} = V_{y,crew} + V_{y,oil} + V_{y,manage} \tag{7}$$

$$V_{y,crew} = \sum_{i \in I_{type}} O_{crew,i} \times N_{yi} \tag{8}$$

$$V_{y,oil} = \sum_{i \in I_{type}} \sum_{rp \in RP} \frac{2F_{y,rp} \cdot M_i \cdot \sum_{b \in W_C} d_b}{N_{y,rpi}} \times 48 \tag{9}$$

$$V_{y,manage} = (V_{y,crew} + V_{y,oil}) \times 10\% \tag{10}$$

$$N_{y,rpi} = \frac{T_{y,rpi} \cdot F_{y,rpi}}{84} \tag{11}$$

In this set of equations, RP is the set of river liners from river ports; $V_{y,crew}$,$V_{y,oil}$, and $V_{y,manage}$ are, respectively, the annual labor, bunker, and management costs incurred by ships operating in the yth year; $O_{crew,i}$ represents the annual labor cost for a ship of size i in the yth year. In Eq. (11), $T_{y,rpi}$ indicates the transportation time of ships of size i on a liner route from river port rp in the yth year; $N_{y,rpi}$ is the number of river ships of size i deployed on a river liner route from river port rp in the yth year; $F_{y,rpi}$ gives the operating frequency (departures/week) of the river liner that operates ships of size i from river port rp. In addition, d_b is the length of river channel link a; W_C is the set of river channel links; M_i (CNY/km) gives the bunker efficiency of river ships of size i.

The lower model, meanwhile, determines the probabilities (P_{y,nr_n}) that one of the feasible routes will be chosen given their service utilities, and calculates the container traffic along routes and links. This operation can be expressed as follows:

$$P_{y,nr_n} = Exp(-\theta C_{y,nr_n}) / \sum_{r_n' \in R_n} Exp(-\theta C_{y,nr_i'}), \quad n \in N \tag{12}$$

$$T_{y,nr_n} = G_{yn} P_{y,nr_n} \tag{13}$$

$$T_{ya} = \sum_{n \in N} \sum_{r_n \in R_n} \delta_{a,nr_n} T_{y,nr_n}, \quad a \in H \cup W_C \cup W_L \cup S \tag{14}$$

Here, T_{y,nr_n} and T_{ya} are the container flows along a given route and link, respectively; θ is a parameter that describes the degree of randomness; C_{y,nr_n} indicates the total cost of transporting containers from origin n to gateway seaport via route r_n in the yth year. In Eq. (13), G_{yn} records the container freight demand from origin n in yth year. In Eq. (14), δ_{a,nr_n} describes whether link a exists on route nr_n. If yes, then $\delta_{a,nr_n} = 1$; otherwise, $\delta_{a,nr_n} = 0$. Finally, H, W_L, and S are the link sets of roadways, inland waterways, and short sea waterways, respectively.

Although the number of possible routes between nodes within a network is infinite, containers will always try to choose the shortest path (that is, the one with the minimum cost). Thus, we assume that the shortest route between the origin and destination nodes n and the gateway seaport is the only feasible route. In order to cooperate with the decision-making output of the upper model (namely, the optimizations of river channel upgrades, river liner ship choice, and operating frequency), we establish three separate port access networks and search for the shortest route using the Dijkstra method. The first access network includes only highway links, and a link's impedance is composed of truck travel time, the cartage fee, and the carbon tax payment. The calculation method is shown below:

$$c_h = d_h \times c_{truck} + \mu \times d_h / v_h + d_h \times f_{hT} \times Coe_{truck} \times Cot, \quad h \in H \tag{15}$$

Here, c_h represents the highway link impedance; d_h is the link length; c_{truck} is the unit price of truck transportation; v_h is the truck's travel speed; μ is the value of time for the container; f_{ht} gives the fuel consumption per unit of trucking distance. Meanwhile, Coe_{truck} represents the CO_2 emissions of trucks powered by fossil fuel, and Cot gives the monetary value of the carbon tax.

The second access network consists of all of the river channel links, the highway links from the origins to their nearest river ports,

and the short seaway links from the ends of inland waterways to the gateway seaport. Eq. (15) can be used to calculate the link impedance for highway transportation from an origin to the nearest river port. The calculation of the impedance (c_b) of a river channel link is shown below in Eq. (16):

$$c_b = d_b \times c_{ship} + \mu \times \frac{d_b}{v_b}\left[1 + \alpha(T_b/Ca_b)^{\beta}\right] + d_b \times \frac{f_{bT}}{T_b} \times Coe_{ship} \times Cot\,, \quad b \in W_C \tag{16}$$

In this equation, c_{ship} and v_b indicate, respectively, the unit transportation cost and speed of a river liner ship; Ca_b gives the capacity of a river channel link, which is determined by the decision variables of river liner ship size (S_{size}) and operating frequency (F_{rp}); f_{ht} is the fuel consumed by a river ship sailing over a given unit of distance. In addition, Coe_{ship} is the emission of a river ship powered by fossil fuel; α and β are the calibrated parameters. The sizes of liner ships that can be purchased and operated are determined by the decision variable of the current river channel grade ($R_{channel}$).

The impedance of river lock (c_b) can then be calculated using Eq. (17):

$$c_b = c_{lock} + \mu \times t_0\left[1 + \alpha(T_b/Cl_b)^{\beta}\right] + \frac{Coe_{lock}}{T_b} \times Cot\,, \quad b \in W_L \tag{17}$$

Here, c_{lock} represents the lock toll; t_0 is the time necessary for a river ship to clear the lock; Cl_b indicates the capacity of the lock link; Coe_{lock} is the CO_2 emitted during the course of one completed lock operation.

The waiting time of containers at a river port can be represented by the impedance of the dummy link that connects a river port terminal to the highway node at the terminal gate, and can be expressed as follows:

$$c_{rp} = c_{R-port} + \mu \times 84/F_{rp}\,, \quad rp \in R_p \tag{18}$$

Here, c_{rp} measures the impedance of the dummy link at river port rp; F_{rp} gives the operating frequency of the river liner from river port rp; R_p represents the set of river ports; $c_{R\ -\ port}$ is the cargo handling fee of river port rp.

The third access network includes all river channel links, all highway links from the origins to their nearest river ports, all links of the short sea routes from feeder ports to the gateway seaport, and the dummy links that connect the highways to river port terminals and the river channels to feeder port terminals. For the third access network, in addition to the impedance functions given in Eqs. (15)–(18), the impedances of the river-sea dummy links and the short sea link, respectively, can be calculated via Eqs. (19) and (20):

$$c_{sp} = c_{S_port} + \mu \times 84/F_{sp}\,, \quad sp \in S_p \tag{19}$$

$$c_b = d_b \times c_{vessel} + \mu \times \frac{d_b}{v_b} + d_b \times \frac{f_{sT}}{T_b} \times Coe_{vessel} \times Cot\,, \quad b \in S \tag{20}$$

In Eq. (19), c_{sp} is the impedance of the river-sea dummy link, and S_p is the set of feeder ports. F_{sp} represents the operating frequency of the short-range sea liners, while f_{sT} is the fuel consumed over a unit of sailing distance by the feeder liner ship. In Eq. (20), Coe_{vessel}

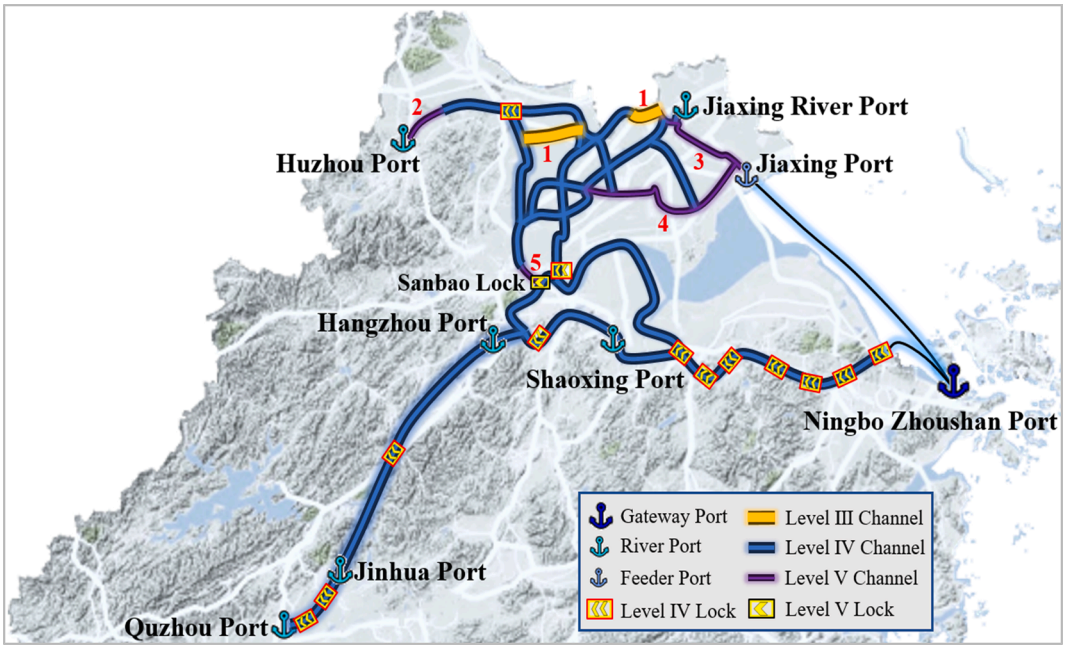


Fig. 3. Current river infrastructure grades and related ports in Zhejiang Province.

indicates the CO_2 emissions of a feeder ship powered by fossil fuels. Meanwhile, S is the set of short-range sea liner routes, and since it is rational to believe that a feeder port operates only one feeder liner route, we assume that $S = S_p$.

As mentioned earlier, the traffic capacity of river channel links and lock links are functions of the decision variables, and thus, the following method can be used to calculate their traffic capacity:

$$Ca_b = S_{size} \times F_{rp}, \quad S_{size} \forall R_{channel}, \quad b \in W_C \tag{21}$$

$$Ca_b = Min\{R_{lock}, \ S_{size} \times F_{rp}\}, \quad b \in W_L \tag{22}$$

Here, S_{size} and $R_{channel}$ indicate the size of the river liner ship and the current river channel grade, respectively. The grade determines the largest ship that can navigate through the channel. In Eq. (22), meanwhile, R_{lock} represents the lock grade (that is, the average daily ship volume), which must be converted into the weekly average passing capacity.

5. Case analysis

For the case study, we take export containers from six origin cities (Hangzhou, Jiaxing, Huzhou, Shaoxing, Jinhua, and Quzhou) located along good inland waterways in Zhejiang Province, and set Chuanshan Container Terminal in Ningbo as the destination. According to *the Medium and Long-term Layout Plan of Inland Waterways and Ports of Zhejiang Province* in 2020, the river shipping network in Zhejiang consists of the Beijing-Hangzhou Canal, the Changhushen Line, the Hangshen Line, the Hujiashen Line, the Zhajiasu Line, the Hangpingshen Line, the Hangyong Canal, the Qiantang River, the Dongzong Line, the Hanghuxin Line, the Meihu Line, the Jiayu Line and 12 additional critical waterways, as shown in Fig. 3.

In China, inland waterways are jointly managed by China's Ministry of Communication (CMOC) and provincial transportation departments, with the provincial transportation departments bearing primary responsibility for the construction and navigation management of provincial waterways. According to the *Navigation Standards for Inland Waterways (GB50139-2014)* compiled by the CMOC, inland waterways can be divided into seven levels according to the tonnage of ships that can be accommodated. Among these, Level III, Level IV, and Level V waterways can accommodate 1000 t, 500 t, and 300 t ships, respectively. The classification of ship locks is also kept consistent with that of the corresponding waterway. The channel levels of the inland waterway in Zhejiang province are depicted in Fig. 3. In Fig. 3, the Hujiashen Line (1) is a Level III channel, while the Meihu Line (2), the Zhajiasu Line (3), the Hangpingshen Line (4), and Beixingqiao-Sanbao lock section (5) of the Beijing-Hangzhou Canal are Level V channels. All other channels and locks are Level IV. Fig. 3 also shows the river ports of Hangzhou, Jiaxing, Huzhou, Shaoxing, Jinhua, and Quzhou, which serve the six container origin cities. Zhapu Terminal in Jiaxing is the feeder sea port for short-range sea liners.

5.1. Data preparation

The Access/Egress Transportation Plan of Ningbo Zhoushan Port has provided a detailed analysis of the future container freight demand expected at Ningbo Zhoushan port, in which the projected container throughput in the years 2025, 2030, 2035, and 2055 reaches 37.21, 43.5, 50.00, and 79.89 million TEU, respectively. Based on these projections of freight demand, we can interpolate the expected container throughput at Ningbo Zhoushan Port in the years of our analysis period (2026–2055). We then apply the average coefficient method to predict the annual container flows from each of the six cities to Ningbo Zhoushan Port based on the container volumes (collected from Customs statistics) reported in 2017. The results of these calculations are tabulated in Appendix 1.

Due to the lack of distribution data for port access and egress transportation flows, it is difficult to calibrate Eq. (12) based on actual observation data. Since the container volume of the different transportation modes for each city is known, however, based on the principles of neuron networks we can repeatedly adjust the value of θ, and use Eq. (12) to calculate probability of cargo owners selecting a given transportation mode at each access and egress port. From this, the container volume of the different transportation modes in each city can be derived. We then take the value of θ (-0.0015) from the calculated result that is closest to the actual data as

Table 1
Values of the input variables.

Variable	Value	Sources
$I_{channel}$	14 *millon CNY/km*	Wang et al. (2022b)
I_{lock}	50 *millon CNY/km*	
α_1	2.75%	https://money.inpai.com.cn/other/20220929/145925.html
θ_2	0.05	https://www.masipo.org.cn/qixian_852766
c_{truck}	5 *CNY/km*	https://www.china-highway.com
v_h	80 *km/h*	https://baijiahao.baidu.com/s?id=1719181476305056083
f_{hT}	0.86 *kg/km*	https://zhidao.baidu.com/question/1576612352906803620/answer/2322600090.html
Coe_{truck}	3.10	https://www.tanjiaoyi.com/article-3075–1.html
Cot	68 *CNY/ton* (2025)	Zhang et al. (2022)
α	0.15	Wang et al. (2022b)
β	4	
Coe_{ship}	3.10	https://www.tanjiaoyi.com/article-3075–1.html
c_{lock}	0 *CNY/ship*	https://3g.163.com/local/article/ENVNOOU804098FEB.html
Coe_{vessel}	3.17	https://www.tanjiaoyi.com/article-3075–1.html

the final value of θ for numerical analysis.

In China, the *Code for Rational Service Life and Durability Design of Water Resources and Hydropower Projects (SL 654–2014)* stipulates that the service life of inland waterways and ship locks is 50 years, and that the net residual value rate of fixed assets is 0. From this information, based on straight-line depreciation, we determine that the value of the annual discount rate, θ_1, of the assets of inland waterways and locks is 0.02. For the purposes of model solution, the values of the other input variables and their sources are shown below in Table 1.

5.2. Solution and analysis

5.2.1. Solution method

In the upper model, the grades of the river channels and locks, river liner ship size, and the operating frequency comprise the decision variables. Since the channel grades both upstream and downstream must be consistent for unimpeded transportation, this implies that the lock grade R_{lock} and the channel grade $R_{channel}$ must match, and that the number of combinations of channel and lock upgrade schemes (R) is limited and can be enumerated. The decision variables of river liner ship size i and operating frequency F_w are discrete variables with few desirable values. For example, subject to the channel grade, only two or three types of ships can be selected, and the frequency of operation ranges from only one voyage/week to ten or more voyages/day. Therefore, all feasible solutions can be enumerated, and the set of the enumerated feasible solutions is $\{R, i, F_w\}_j \in J$.

Highway infrastructure, meanwhile, is not dedicated solely to container trucking, but rather serves various forms of transportation, and has mainly been planned and constructed by governments in response to social demand. Thus, a container truck's travel time does not change with its flow along a highway link, but with total traffic flow along that link; furthermore, the freight charge should be equivalent to the equilibrium price of participating in the highway transportation market. Truck transportation time and the attendant freight charge can thus be treated as exogenous variables. At river channels and locks, in which port access liner flows represent the main source of traffic, it is rational to suppose that ship sailing time varies with river liner traffic flows. Moreover, river liner freight charges are affected by both ship size and operating frequency. These factors indicate that the transportation impedance of a river shipping link and the probability of a shipper choosing to transport a container via river liner interact with each other. In order to obtain the decision variables (including the plan of river channel upgrades, river liner size, and operating frequency) under conditions of an equilibrium between supply and demand for river liner service, this study designs a solution algorithm for the lower model based on the Frank-Wolfe method. The steps are as follows:

Step 0: Input the value of $Ca_b(b \in W_L)$,$F_w(w \in R_p \cup S_p)$ and ship size i, which are the outputs from the upper model;

Step 1: Let $k = 0$, calculating the impedances of $c_h^k, c_b^k, c_{rp}^k, c_{sp}^k, c_s^k$ under the condition of zero flows along waterway links;

Step 2: Search for the shortest route between the origin and the destination, respectively, on the three port access networks using the Dijkstra algorithm in order to obtain the impedance $C_{nr_n}^k(n \in N)$ of the shortest paths;

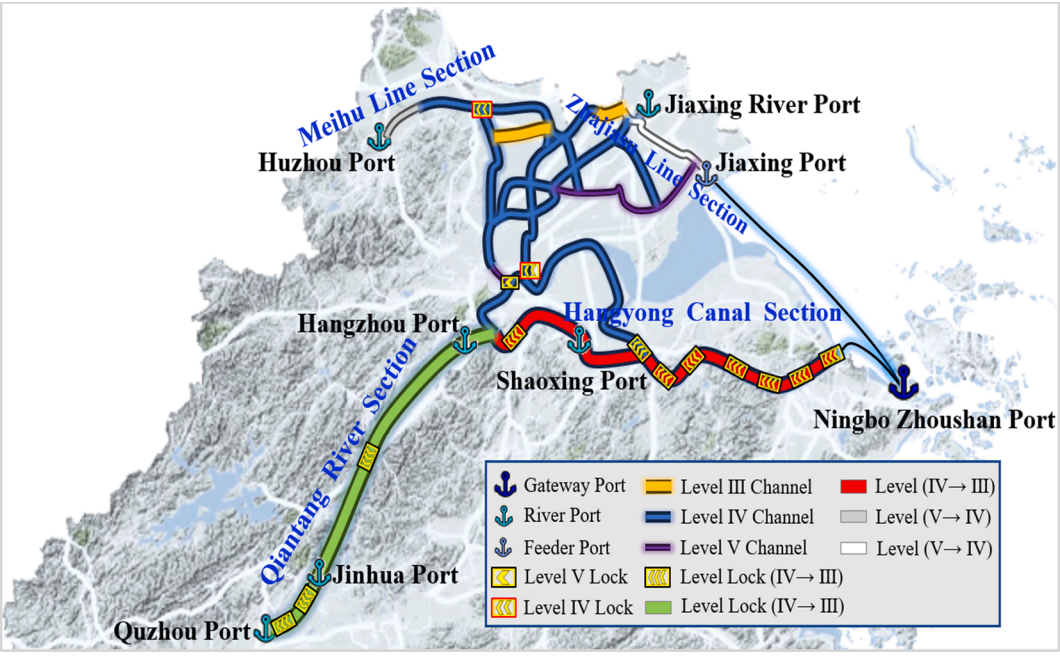


Fig. 4. The optimal river channel upgrade scheme.

Step 3: Use Eqs. (12)–(14) to calculate the selection probability ($P_{nr_n}^k$) of each shortest route, the route's container traffic $T_{nr_n}^k$, and the link' s container traffic T_a^k;

Step 4: Calculate the impedance of $c_h^{k+1}, c_b^{k+1}, c_{rp}^{k+1}, c_{sp}^{k+1}, c_s^{k+1}$, given waterway links with traffic flows of T_a^k;

Step 5: Let $k = k + 1$, and update link traffic as follows: $T_a^k = T_a^{k-1} + \lambda(T_a^{k-1} + T_a^{k-2})$;

Step 6: Determine whether to terminate the calculation. If $\left|T_a^{k-1} - T_a^{k-2}\right| \langle \varepsilon$ (ε is a predetermined threshold), stop the calculation; otherwise, return to Step 2.

5.2.2. Solution analysis

Using MATLAB 2014a to design the solution program and solve the model, we have determined a river channel upgrade scheme (shown in Fig. 4), the appropriate size of river ships and the frequency at which they should operate (shown in Fig. 5), and the likeliest travel routes of containers transported by river liner from the six origin cities to port with the minimum total generalized cost during the payback period (2026–2055).

In Fig. 4, the optimal approach to infrastructure improvements is to upgrade the Qiantang River and Hangzhou-Ningbo Canal channels from Level IV to Level III, and the Meihu Line and Zhajiasu Line channels from Level V to Level IV. Qiantang River and Hangzhou-Ningbo Canal are the backbone of the inland waterway system in Zhejiang Province. The freight demand in the hinterlands of Hangzhou, Shaoxing, and Jinhua are each high, and so this output aligns with reality. With the improvements to the river channels, river liner companies can use larger ships to carry more containers and further realize critical economies of scale. The current grade of the river channels along the Meihu Line and Zhajiasu Line is Level V, which currently does not allow them to accommodate river liners. From the model solution, it is thus clear that in order to operate river liners from Huzhou and Jiaxing to the Zhapu Terminal (the feeder sea port in this case study), it is necessary to upgrade the river channels of Meihu Line and Zhajiasu Line to grade IV.

As depicted in Fig. 5, after the river channel upgrades of Qiantang River and Hangzhou-Ningbo Canal to Level III, the river liner companies operating out of Hangzhou Port, Shaoxing Port, Jinhua Port, and Quzhou Port will all choose larger vessels (96 TEU). This is because the river channels now allow a ship of 96 TEU to travel from Hangzhou Port, Shaoxing Port, Jinhua Port, or Quzhou Port all the way to the river estuary, while the freight demand of the these ports' hinterlands is large enough to fully utilize the extra shipping capacity. Meanwhile, if the river channels of Meihu Line and Zhajiasu Line are upgraded to Level IV, the river liners originating from Jiaxing River Port and Huzhou Port will still operate small ships (36 TEU). This is primarily because the upgraded channels between these two ports and the Zhapu Terminal (feeder port) can still only accommodate small ships. In these areas, however, the projected annual increases in container freight demand can be adquately served by increasing the operating frequency of these smaller river liners.

Finally, Fig. 6 demonstrates the two main corridors of river shipping for container port access transportation in Zhejiang Province. These two paths are formed by the two shortest river shipping routes from the six origin cities to Chuanshan Container Terminal. Of these, Corridor 1 begins at the middle and upper reaches of Qiantang River (Quzhou, Jinhua, western Hangzhou), passes along the Qiantang River and Hangzhou-Ningbo Canal, and finally reaches Chuanshan Terminal. Corridor 2 originates from the Huzhou and Jiaxing areas, traverses the Changhushen Line, Zhajiasu Line, and Jiaxing Branch Lines, before reaching Chuanshan Terminal.

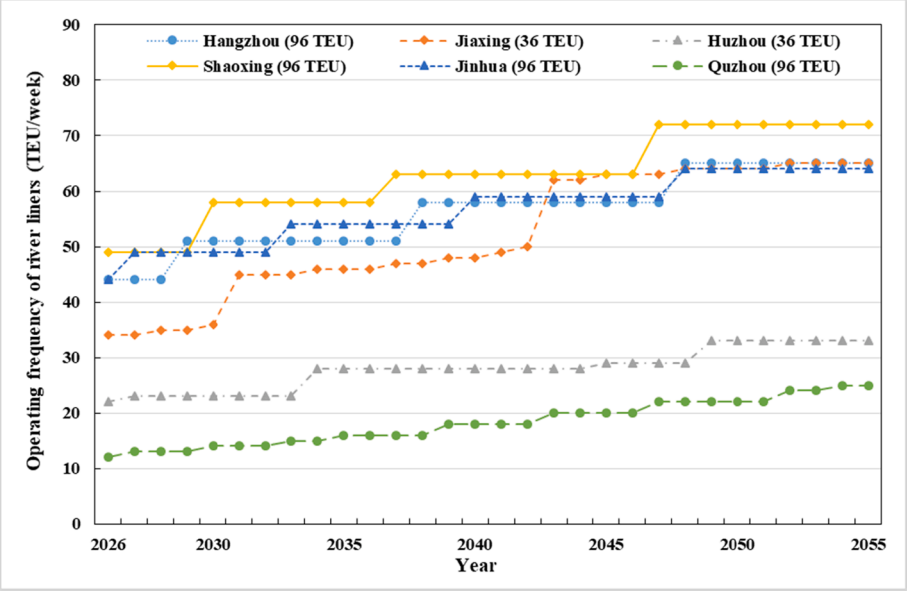


Fig. 5. Optimal ship size and operating frequency of river liners.

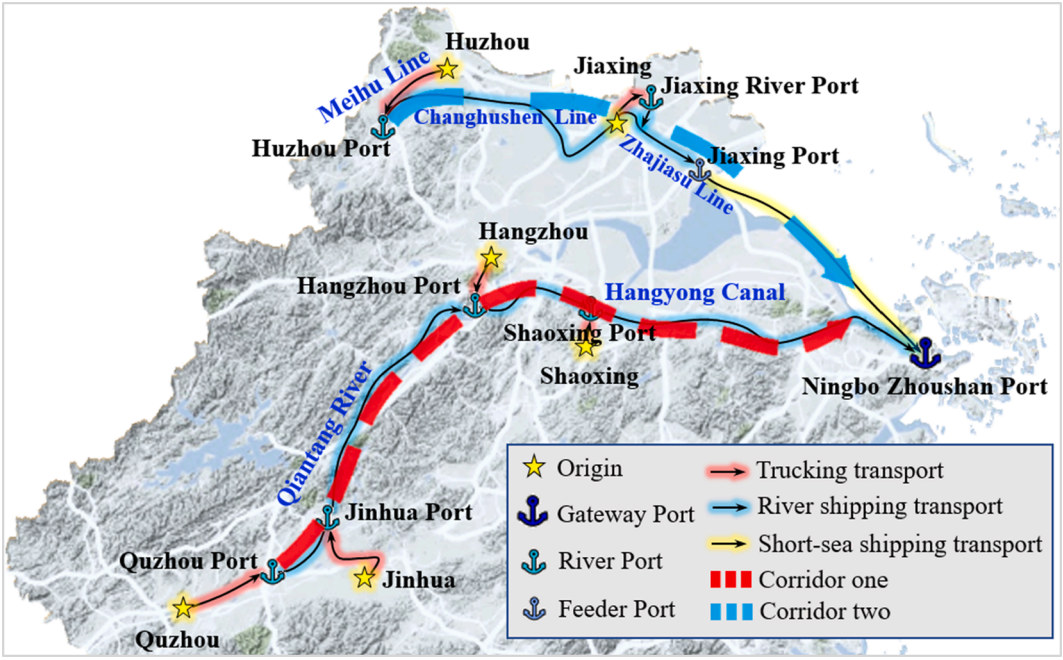


Fig. 6. The two main river shipping corridors for port access transportation.

5.2.3. Analysis of mode shares

Given the projected costs of trucking and river shipping to gateway container ports over the next 30 years, we can calculate the annual mode shares of trucking and river shipping from each of the six cities, as shown in Fig. 7.

In Fig. 7, the mode share of river shipping from all cities is projected to increase slowly year by year. This is because the transit and waiting time of containers at river ports near the origin nodes will decrease as the operating frequency of river liners increases, which reduces the overall time cost of river shipping to container ports. Of all the cities, Jiaxing is projected to have the largest mode share of river shipping, reaching 38.8% by 2055. This is mainly due to the fact that along the route from Jiaxing River Port to Zhapu Terminal (and thence to Chuanshan Terminal), the river leg is shorter than the short-range sea feeder leg, which allows containers along this route to take advantage of the significant economies of scale of larger oceangoing ships. Huzhou is projected to have the smallest mode share of river shipping to Chuanshan Terminal, at only 25% by 2055. This can be explained by the fact that river shipping from Huzhou to Chuanshan Terminal can only use small ships (36 TEUs), and thus the economies of scale remain underdeveloped. Moreover, the transition of containers at Zhapu Terminal may actually further increase the port access cost to shippers from Huzhou. The mode share

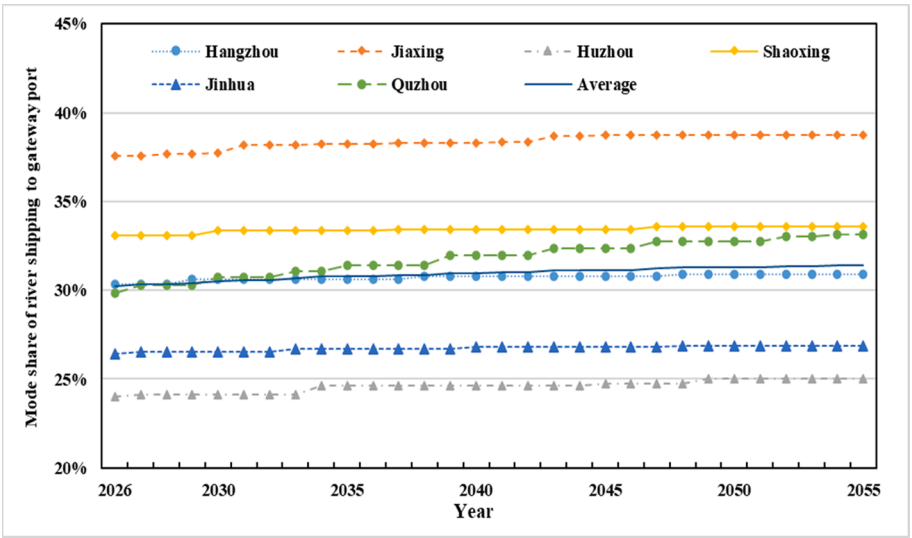


Fig. 7. Projected mode share of river shipping per city over 30 years.

of river shipping from Quzhou is expected to rise from 29.8% in 2026 to 33.2% in 2055, demonstrating the most significant rate of increase. Although the river lines operating from Quzhou to Chuanshan Terminal use large capacity ships (96 TEUs) that have a low rate of energy consumption per container, in the early stages the small volume of freight demand restricts the operating frequency of river liners at Quzhou Port and increases container waiting times. As freight demand continues to increase, the operating frequency of river liners at Quzhou port will also gradually increase and reduce these time costs, thus increasing the mode share of river liner shipping to gateway ports significantly at Quzhou.

5.2.4. Carbon emission analysis

Based on the projected mode shares of container port access transportation over the study period, the carbon emissions of trucking and river shipping from the six cities can now be estimated, and the volume of emissions both with and without river channel upgrades can be compared. Fig. 8 tabulates the calculation results below. For each of the two test cases, the total emissions of CO_2 in 2026 are projected to be 1.4206 and 1.9255 million tons, respectively, illustrating that channel upgrades will promote a shift in the mode of container port access transportation from trucking to river shipping. This implies a reduction in carbon emissions from container port access transportation of approximately 0.505 million tons in 2026 (a decrease of 26.2%). In 2030, 2035, 2040, 2045, 2050, and 2055, CO_2 emissions are expected to decrease by 0.57 (26.44%), 0.665 (26.56%), 0.768 (26.69%), 0.869 (26.73%), 0.974 (26.86%), and 1.075 million tons (26.88%), respectively. If the river channels are upgraded, total CO_2 emissions from the container port access transportation of the six cities across the entire repayment period (30 years) is projected to be 64.43 million tons, or 23.45 million tons less (a decrease of 26.68%) than when no river channel upgrades are made. Upgrades to the river channels are thus likely to both increase the mode share of river liner shipping and to effectively reduce carbon emissions resulting from container port access transportation. Moreover, these results suggest that the effect will become increasingly obvious as container freight demand continues to grow.

5.3. Sensitivity analysis of a carbon tax

The results from Section 5.2.2 are obtained assuming a situation in which $Cot = 68$ CNY/ton. As the pressure increases to reduce emissions, one commonly prescribed method of control is the increase or implementation of a carbon tax. Thus, it is necessary to conduct a sensitivity analysis of changes in the carbon tax rate. The findings of Zhang et al. (2022) suggest that the carbon tax rates in China are likely to be set at 68, 178, 435, or 1,363 CNY/ton during the period 2020–2060. We use these suggestions as the basis for the sensitivity analysis of changes in the carbon tax rate, and the results are summarized below in Fig. 9.

Over the next 30 years, in the absence of a carbon tax (0 CNY/ton), the mode share of river shipping as a component of port access transportation remains at 29.03%, while the generalized cost of port access transportation, and carbon emissions total are calculated at 178.64 billion CNY and 65.52 million tons, respectively. When the carbon tax rate is raised to 68 CNY/ton of carbon, the mode share of river shipping and the cost of port access transportation will increase, respectively, by 0.013 and 3.841 billion CNY. Simultaneously, the total cost (excluding the carbon tax payment) and the level of carbon emissions will each be reduced by 540 million CNY (0.30%) and 1.083 million tons (1.65%), respectively. When the tax rate increases again to 178 CNY/ton of carbon, the mode share of river shipping and the transportation cost will increase, respectively, by 0.035 and 9.7 billion CNY. Meanwhile, the cost without the carbon tax payment and the amount of carbon emissions will each decrease by 1.447 billion CNY (0.81%) and 2.898 million tons (4.42%),

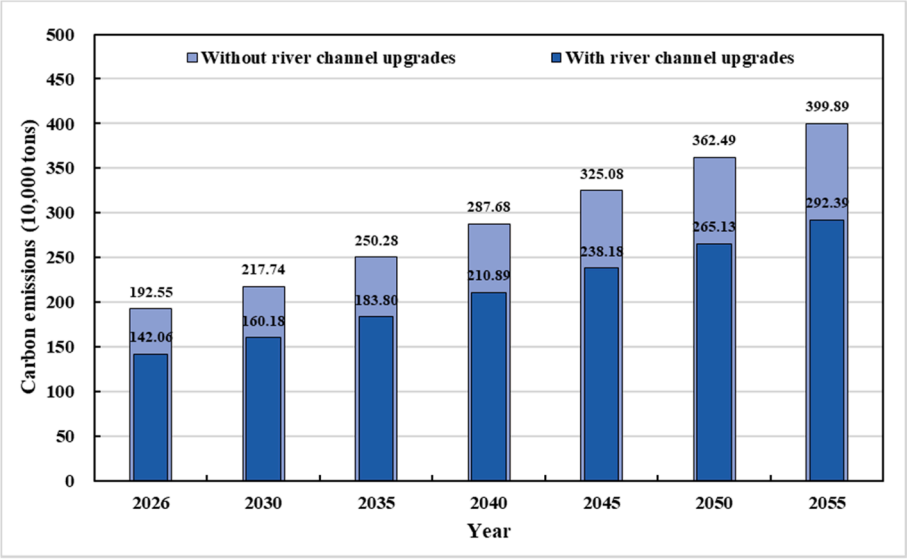


Fig. 8. The impact of channel upgrades on CO_2 emissions from port access transportation.

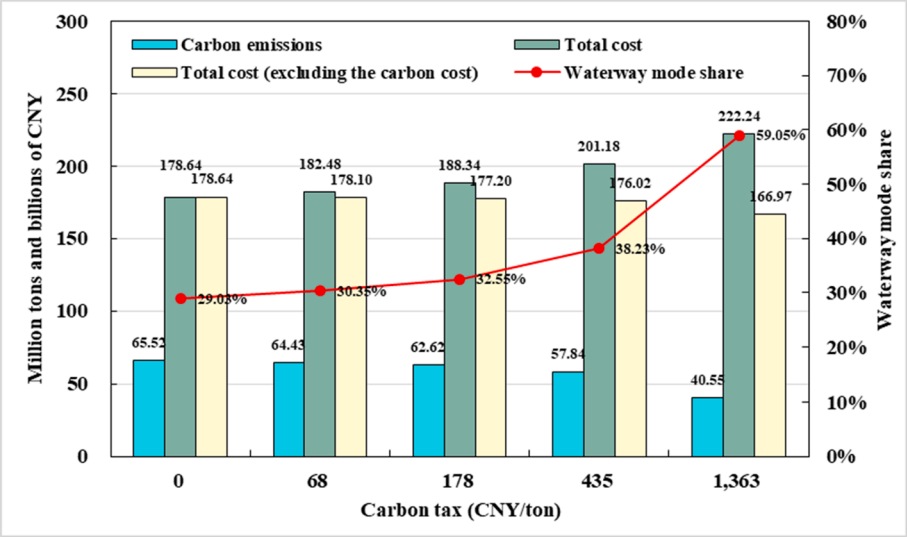


Fig. 9. Mode shares, CO_2 emissions, and transportation costs at five different tax rates.

respectively. When the tax rate rises once again to 435 CNY/ton of carbon, the mode share of waterway shipping and the transportation cost will increase, respectively, by 0.092 and 22.54 billion CNY. In addition, the cost excluding the carbon tax payment and the level of carbon emissions will again decrease by 2.617 billion CNY (1.46%) and 7.68 million tons (11.65%), respectively. In the final case, when the tax rate is raised to 1,363 CNY/ton of carbon, the mode share of river shipping and the transportation cost will once again increase, respectively, by 0.3 and 43.60 billion CNY. Finally, the transportation cost (excluding the carbon tax payment) and the level of carbon emission will decrease by 11.671 billion CNY (6.53%) and 24.97 million tons (38.11%), respectively.

The above analysis demonstrates that as the carbon tax rate increases, both the mode share of river shipping and the total generalized cost of port access transportation will rise. Simultaneously, the transportation cost (minus the carbon tax payment) and the level of carbon emissions will both decrease. In both cases, the margins of the increases and decreases in these measures expand as the increment change in the carbon tax rate becomes larger. This is primarily because trucking is more sensitive to a carbon tax than river shipping, and a high carbon tax will thus encourage more containers to shift from trucking to river shipping. As a result, the CO_2 emitted from container port access transportation is projected to decrease. Because the savings in the carbon tax payment due to the mode shift from trucking to river shipping is smaller than the overall increase in the rate of the carbon tax, however, the total generalized cost of transportation will continue to rise. Meanwhile, as the tax rate increases lead to a steady rise in the mode share of river shipping, the economies of scale from river shipping are likely to grow more pronounced due to the increase in freight demand. This development is likely to lead to further reductions in the transportation cost (excluding carbon costs) for container port access shipments.

As the carbon tax rate rises, the plans for river channel upgrades will also have to change. The recommendations in the case when the carbon tax is equal to 68 CNY/ton remain exactly the same as when the carbon tax is equal to either 0 CNY/ton or 178 CNY/ton. If the carbon tax rises to 435 CNY/ton or 1,363 CNY/ton, however, the optimal upgrade plans for river channels will change. Most significantly, the river channels from Huzhou Port to the sea (Levels IV and V) and from Jiaxing River Port to the sea (Levels IV and V) will all be upgraded to Level III. This is primarily due to the pronounced shift from trucking to river transportation as a result of increases in the carbon tax rate, which produces further economies of scale and income benefits from waterway shipping. The return on the investment in upgrading all of these river channels to Level III is now projected to be greater than the investment costs. In addition, it is worth noting that, despite the channel upgrades, only the river line from Huzhou Port to Chuanshan Terminal will shift to large ships (96 TEU), while the river lines from Jiaxing River Port to Zhapu Terminal will continue to rely on small ships (36 TEU). This is because the river liner voyage from Huzhou Port to Zhapu Terminal is relatively long, rendering the economies of scale from large ship capacity significant. The freight demand in the Jiaxing area, meanwhile, is projected to remain relatively low and the river liner voyage from Jiaxing River Port to Zhapu Terminal is also short, thus reducing the economies of scale from large vessels.

6. Conclusion

At present, the mode share of river shipping as a component of China's container port access and egress transportation remains small. Except along the golden waterway of the Yangtze River, the growth of river shipping even in regions rich in river channels has lagged behind expectations. This study builds a bi-level programming model in order to provide suggestions for improving river channels, increasing the mode share of river shipping in container port access and egress transportation, and controlling the corresponding carbon emissions. The model carries out these tasks with the objective of minimizing the sum of the carbon tax payment, investment costs in river channel upgrades and ship purchases, the costs of ship operation, and the time and money costs of container

transportation from an origin city to a gateway seaport. The upper model determines the optimal plan for river channel upgrades, ship size, and operating frequency based on official projections of container freight demand over the next 30 years. The lower model, meanwhile, assigns containers to the likeliest routes from their origin to the gateway seaport.

The empirical analysis is conducted on data collected from six cities in China's Zhejiang Province. The numerical results show that, first, it is best to upgrade the Qiantang River and Hangzhou-Ningbo Canal channels from Level IV to Level III, and the Meihu Line and Zhajiasu Line channels from Level V to Level IV. Second, due to the channel upgrades, river shipping lines from Hangzhou Port, Shaoxing Port, Jinhua Port, and Quzhou Port should shift to larger ships (96 TEU), while those operating from Jiaxing River Port and Huzhou Port should continue to rely on small ships (36 TEU). Third, if the river channels are upgraded, the mode share of river shipping from the six cities is projected to increase year over year across the 30 years of the repayment period of the investment, until it stabilizes at approximately 30% of container port access transportation. Fourth, upgrading the river channels is expected to effectively reduce carbon emissions from container port access transportation, and these reductions will become more pronounced as container freight demand continues to grow. Fifth, if the carbon tax rate increases from 0 to 68, 178, 435, and finally 1,363 CNY/ton, the CO_2 emitted from container port access transportation will decrease by 1.0827, 2.8976, 7.6806, and 24.9676 million tons, respectively, over the 30 year period. Simultaneously, the mode share of river shipping is expected to increase by 0.013, 0.035, 0.092, and 0.302, respectively. Finally, when the carbon tax rate is more than 435 CNY/ton, the optimal upgrade scheme will change such that the river channels from Huzhou Port to the sea (Levels IV and V) and from Jiaxing River Port to the sea (Levels IV and V) will also receive upgrades to Level III.

The above conclusions provide a preliminary foundation for the further development of river shipping as an element of container port access transportation in China, and for further improvements in the quality of river shipping service. When constructing the programming model, however, this paper does not consider two key issues: the increment change in impedance along river channels due to ship-to-ship transfers as a result in differences in river grades; and, the differences in the amount of channel upgrade investment necessary due to variations in geography and environment. Moreover, this paper only considers port access transportation but ignores the case of egress transportation. This is because China is an export-oriented economy, and the statistics indicate that for every four heavy containers exported from Zhejiang, only one heavy container will be imported. Thus, we have chosen to emphasize the case of export containers. In addition, the time windows for egress transportation of import containers are very flexible. Many are delivered to locations outside the port terminal via backhaul trips of port access trucks or other types of vehicles. It is thus difficult to incorporate them into the model. In ignoring egress transportation, our model may underestimate the effect of upgrades along river channels; however, we believe that decisions made on the basis of our results will have a more solid foundation.

According to the Asian Development Bank (2016), the imbalance between import and export containers will make it difficult to provide more frequent inland liner service, as the increase in frequency also raises the empty load rate of returning liners. This phenomenon is similarly reflected in truck's container port access and egress transportation. Due to the higher shipping costs and carbon emissions of a single container transported by truck, the marginal loss of more empty loads due to the increased frequency of river liner service is far lower than the marginal benefit gained by reducing the number of return container trucks. Therefore, in an area with a robust inland waterway transportation network but a high mode share of trucking in container port access and egress transportation, more frequent river liner services to promote a modal shift from highway to river shipping can not only reduce the cost and carbon emissions of container access and egress around ports, but also reduce the overall waste caused by empty vehicle load return.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix 1

	Ningbo Zhoushan Port (Unit: 10,000 TEUs)									
	2026	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**
Hangzhou	86.26	89.09	91.91	94.73	97.55	100.46	103.38	106.29	109.21	112.13
Jiaxing	14.38	14.85	15.32	15.79	16.26	16.74	17.23	17.72	18.20	18.69
Huzhou	13.27	13.71	14.14	14.57	15.01	15.46	15.90	16.35	16.80	17.25
Shaoxing	91.79	94.80	97.80	100.80	103.80	106.90	110.01	113.11	116.21	119.31

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(*continued*)

	Ningbo Zhoushan Port (Unit: 10,000 TEUs)									
	2026	**2027**	**2028**	**2029**	**2030**	**2031**	**2032**	**2033**	**2034**	**2035**
Jinhua	106.17	109.64	113.12	116.59	120.06	123.65	127.24	130.82	134.41	138.00
Quzhou	12.17	12.56	12.96	13.36	13.76	14.17	14.58	14.99	15.40	15.81
Total	324.04	334.65	345.25	355.84	366.44	377.38	388.34	399.28	410.23	421.19

	Ningbo Zhoushan Port (Unit: 10,000 TEUs)									
	2036	**2037**	**2038**	**2039**	**2040**	**2041**	**2042**	**2043**	**2044**	**2045**
Hangzhou	115.48	118.83	122.18	125.53	128.88	132.23	135.58	138.94	142.29	145.64
Jiaxing	19.25	19.80	20.36	20.92	21.48	22.04	22.60	23.16	23.71	24.27
Huzhou	17.77	18.28	18.80	19.31	19.83	20.34	20.86	21.37	21.89	22.41
Shaoxing	122.88	126.45	130.01	133.58	137.14	140.71	144.28	147.84	151.41	154.98
Jinhua	142.12	146.25	150.37	154.50	158.62	162.75	166.87	171.00	175.12	179.25
Quzhou	16.29	16.76	17.23	17.70	18.18	18.65	19.12	19.59	20.07	20.54
Total	433.79	446.37	458.95	471.54	484.13	496.72	509.31	521.90	534.49	547.09

	Ningbo Zhoushan Port (Unit: 10,000 TEUs)									
	2046	**2047**	**2048**	**2049**	**2050**	**2051**	**2052**	**2053**	**2054**	**2055**
Hangzhou	148.99	152.34	155.69	159.04	162.40	165.75	169.10	172.45	175.80	179.15
Jiaxing	24.83	25.39	25.95	26.51	27.07	27.62	28.18	28.74	29.30	29.86
Huzhou	22.92	23.44	23.95	24.47	24.98	25.50	26.02	26.53	27.05	27.56
Shaoxing	158.54	162.11	165.67	169.24	172.81	176.37	179.94	183.51	187.07	190.64
Jinhua	183.37	187.50	191.62	195.75	199.87	204.00	208.12	212.25	216.37	220.50
Quzhou	21.01	21.48	21.96	22.43	22.90	23.37	23.85	24.32	24.79	25.27
Total	559.66	572.26	584.84	597.44	610.03	622.61	635.21	647.80	660.38	672.98

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